

IUDX Data Access Request - Neer Vazhvu

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Project: Neer Vazhvu - Chennai Water Intelligence Dashboard

Website: <https://neer-vazhvu.vercel.app> **Source Code:** <https://github.com/SundareshPrasanna/neer-vazhvu> (open-source, MIT license)

Project Overview

Neer Vazhvu is an open-source civic dashboard that visualizes Chennai's water infrastructure - reservoirs, groundwater, rivers, flood risk, and water bodies - to help citizens, journalists, and policymakers understand the city's water health.

The platform currently integrates data from CPCB (river water quality), CGWB/India WRIS (groundwater), IMD (rainfall), NASA POWER (climate), OpenStreetMap (water body geometry), and OpenCity Chennai (flood hazard zones).

Requested Dataset

Flood Level Alerts from Flood Monitoring Systems in Chennai City Resource Group ID: 257aab1b-1258-445a-a37e-058486a2fa13

Purpose of Use

We intend to integrate real-time water level sensor readings into the existing flood risk map on our dashboard. Specifically:

1. **Flood risk visualization** - Overlay live canal, subway, and river water levels on the flood hazard map so citizens can see current waterlogging conditions in their area.
2. **Subway/underpass safety** - Surface real-time water levels at 27 subway sensors to warn commuters about flooded underpasses during rain events.
3. **Lake and canal monitoring** - Display current water levels alongside historical flood zone data to provide context during monsoon season.

How the Data Will Be Used

- Data will be fetched via the IUDX Resource Server API and cached for 5 minutes server-side.
- Sensor readings will be displayed as map markers color-coded by water level severity (normal/warning/danger).
- All data will be attributed to "IUDX / Chennai Smart City" on the dashboard.
- No data will be stored permanently or redistributed - only real-time display.
- The dashboard is freely accessible to the public with no login required.

Impact

Chennai faces recurring urban flooding (2015, 2023, 2024 Cyclone Michaung). Real-time water level data, combined with our existing flood hazard zones and drainage network maps, can help citizens make informed decisions during rain events - which roads to avoid, which subways are flooding, and which areas are at immediate risk.

Duration

12 months (with renewal request if continued access is needed).

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